GROUP 2608 - Comparative Analysis of GBDTs and Neural Architectures

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This paper presents a comparative analysis of three Gradient Boosting Decision Tree frameworks (XGBoost, LightGBM, CatBoost) and domain-tailored Neural Networks (CNNs, LSTMs) across structured, spatial, and sequential data regimes. We evaluate hyperparameter sensitivity, noise robustness, computational efficiency, and the effect of data representation on predictive performance. On low-dimensional tabular data, all models converge to comparable accuracy ceilings, with GBDTs exhibiting greater computational efficiency and stronger resilience to feature corruption. On image and time-series data, GBDTs degrade severely without spatial or temporal biases, though hybrid CNN-GBDT pipelines partially recover this deficit. Notably, raw signal flattening consistently outperforms manual feature engineering for tree-based classifiers on sequential tasks. Finally, we explore the feasibility of integrating a GBDT as a GAN discriminator via a Reinforcement Learning framework, demonstrating that mode collapse can be mitigated through explicit regularization. All results are dataset-specific and should not be interpreted as general operational boundaries.

INTRODUCTION

Gradient Boosting Decision Trees (GBDTs) and Neural Networks (NNs) dominate predictive data analysis, yet excel in fundamentally different regimes. GBDTs remain the standard for structured, tabular data by establishing discrete, axis-aligned decision boundaries on permutated features. Conversely, deep learning architectures, such as Convolutional and Recurrent Neural Networks (CNNs, RNNs), dominate unstructured domains by imposing hard constraints (weight sharing, spatial locality, sequential causality) to explicitly exploit data geometry.

Beyond raw accuracy, practical deployment demands rigorous evaluation of model robustness and scalability against feature degradation, label corruption, and high-dimensional collinearity. To evaluate these operational boundaries, we compare three state-of-the-art GBDTs (XGBoost, LightGBM, CatBoost) against domain-tailored NNs across five systematic phases: (1) **Hyperparameter Sensitivity and Baseline Evaluation** and (2) **Robustness Under Controlled Degradation** (feature noise and targeted feature noise); (3) **Scalability in High Dimensional Spaces and Spatial Locality and Image Classification**; (4) **Temporal Dependencies, Sequential Forecasting and Multi-Classification**; and (5) **Generative models based on boosting**. This work aims to provide a clear roadmap for model selection constrained by data quality and geometric complexity. Code / contibution available at [1].

METHODS

To evaluate model performance across diverse tasks, we utilize five primary datasets. The **MAGIC Gamma Telescope dataset** [2] ($\sim 1.9 \times 10^4$ samples, 10 features) serves as the baseline for tabular binary classification and targeted noise degradation tasks. The **MNIST database** [3] tests spatial locality under morphological perturbations, including rotation, translation, occlusion, and contrast shifts. The **Human Activity Recognition (HAR) dataset** [4], containing 3D accelerometer and gyroscope signals, evaluates sequential forecasting and multiclass decision boundaries. Finally, the **IRIS dataset** [5] is a tabular dataset used for taxonomy and classification, containing 150 instances of iris flowers. Each record features four numerical measurements in centimeters: sepal length, sepal width, petal length, and petal width.

We selected three dominant GBDT frameworks to evaluate their structural inductive biases against established deep learning baselines. For the tree based algorithms:

- **XGBoost** [6] employs a second order Taylor expansion of the loss function, capturing enhanced curvature information for gradient descent while integrating L1/L2 regularization directly into the gain calculation via sparsity aware, level-wise tree growth.

- **LightGBM** [7] utilizes a leaf wise growth strategy optimized for speed and large scale data, implementing Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to aggressively compress sparse, high dimensional data without significant information loss.

- **CatBoost** [8] leverages symmetric oblivious trees, applying a uniform splitting criterion across entire tree levels. This structure provides a strong regularizing inductive bias and highly optimized inference speeds, while ordered boosting effectively mitigates prediction shift.

For the deep learning baselines, we deployed a **Convolutional Neural Network (CNN)** to evaluate spatial translation equivariance, and a **Long Short-Term Memory (LSTM)** network [9] to process sequential dependencies without suffering from vanishing gradients.

Neural network stability and generalization were enforced via Batch Normalization [10] and Gaussian noise data augmentation.

All models were subjected to rigorous hyperparameter tuning to establish optimized baselines before degradation testing. For standard tabular data (MAGIC) and high dimensional scaling, we utilized the Bayesian Tree Based Parzen Estimator (TPE) algorithm via the Optuna framework [11]. For the temporal and sequential tasks (HAR), hyperparameter grid searches utilized `HalvingRandomSearchCV` [12] and early stopping [13] to ensure computational efficiency across the expansive search space.

Performance evaluation was tailored to the specific task architecture. Binary classification and noise degradation tasks were evaluated using the Receiver Operating Characteristic Area Under the Curve (ROC-AUC) [14] to sensibly penalize false positives, optimized against a standard 0-1 loss function. Multiclass temporal tasks were strictly assessed via Macro-F1 scores to ensure performance consistency across balanced activity classes. To assess architectural robustness, baseline performances were subsequently tested against controlled degradation, including targeted Gaussian noise injections and adversarial label flipping. High-dimensional scalability was measured by tracking latency across expanding feature sub-spaces. Finally, relative feature and hyperparameter importance were approximated using SHapley Additive exPlanations (SHAP) [15] and functional Analysis of Variance (fANOVA) [16], respectively.

RESULTS

1 - Hyperparameter Sensitivity and Baseline Evaluation In the first two experiments, we study performances of the decision tree-based boosted models versus a Deep (4 hidden layers) Neural Network on the MAGIC dataset, a benchmark Machine learning binary classification task. Each of the $\sim 1.9 \cdot 10^4$ Monte Carlo generated samples consists of 10 geometrical features, calculated from an individual preprocessed image of a hadronic "shower" recorded by a ground based telescope, which models must classify as either caused by a high-energy gamma ray (signal) or a generic cosmic ray (background) colliding with particles in Earth's upper atmosphere. We use a Random forest (RF) and a GradientBoosting classifiers in `SciKitLearn` [12] as baselines. As a validation and testing (non-differentiable) metric we maximise the Receiver Operator Characteristic Area Under Curve (ROC-AUC) in order to sensibly penalize false positives more than false negatives; as a training metric, we minimise a simple 0-1 loss function (error rate). We employ early stopping at all times. Firstly, we tune hyperparameters of each of the five models using the Bayesian Tree Based Parzen Estimator (TPE) search algorithm [17] implemented in the pythonic Optuna module [11]. We use the hyperparameters found to lead to the best (validation) ROC-AUC in all subsequent MAGIC experiments. For each model, we approximate hyperparameter relative importance using functional ANalysis Of VAriance (fANOVA) [16, 18], and the relative importance of the dataset's features with SHapley Additive exPlanations (SHAP) [15, 19] scores. All computations are performed on a Colab Virtual Machine [20], leveraging its T4 GPU whenever possible.

Optuna's [11] hyperparameter optimization's history over only 50 trials (RF was simply initialized 50 times) per model shows fluctuations in a relatively narrow range of good validation accuracies ($\text{ROC-AUC}_t \in]0.88, 0.94[$) similarly for all models (baselines, boosted, and NN); the final accuracy of each model also varies in a narrow range, $\text{ROC-AUC}_{final} \in]0.940, 0.945[$ (Figure 1). The most important (fANOVA [16, 18]) hyperparameters for boosted models (Figure 2) in this context seem to be learning rate, followed by maximum tree depth and number of estimators used; for XGBoost, the number of samples observed by each tree and minimum and minimum number of samples in each split (both parameters preventing overfitting) are also meaningful. These results align with general expectations for this kind of tasks. Overall, we observe how a relatively simple dataset such as MAGIC turns out to be an easy task for these classifiers, which all achieve similar high accuracy with minimal hyperparameter optimization. In terms of computational time, however, GBM is very demanding, as expected from a non GPU optimized basic boosted model; surprisingly,

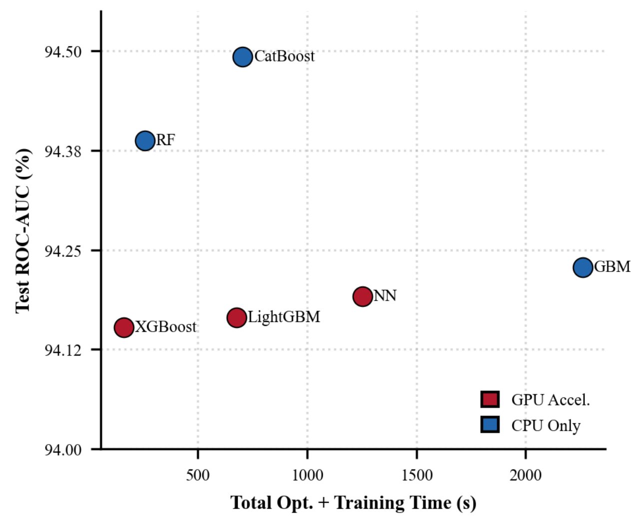


FIG. 1: Model performance (test ROC-AUC) versus total optimization and training time on the MAGIC dataset. Each point represents one classifier; red points mark models that were tuned or trained at least in part with GPU.

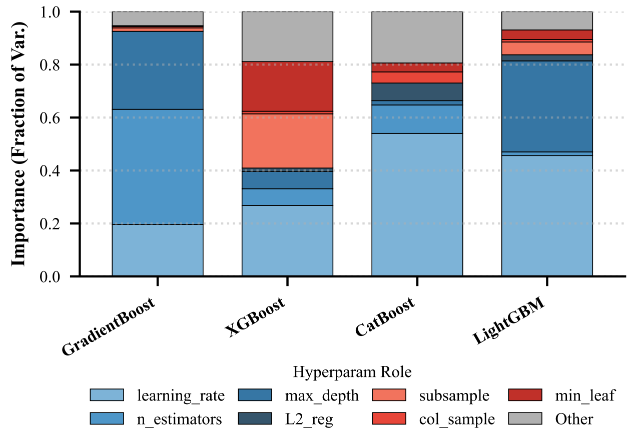


FIG. 2: Relative hyperparameter importance, estimated via functional ANOVA (fANOVA) [16, 18] in Optuna [11], for the four boosted models on the MAGIC dataset.

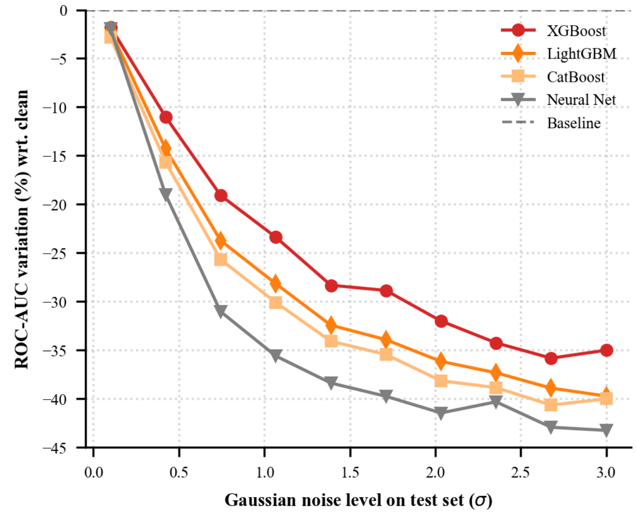
LightGBM also performs similarly to others and worst than XGBoost, which we blame again on the dataset being too small and simple to leverage this model's capabilities. We also note that each model's exact performance varies only slightly across different runs of the code, signaling once again that an accuracy ceiling is almost always achieved on this dataset.

2 - Robustness Under Controlled Degradation. To assess model robustness, we then inject random Gaussian noise of varying magnitudes scaled by the dataset's standard deviation for each feature, σ independently for each feature into the test set (Figure 3a).

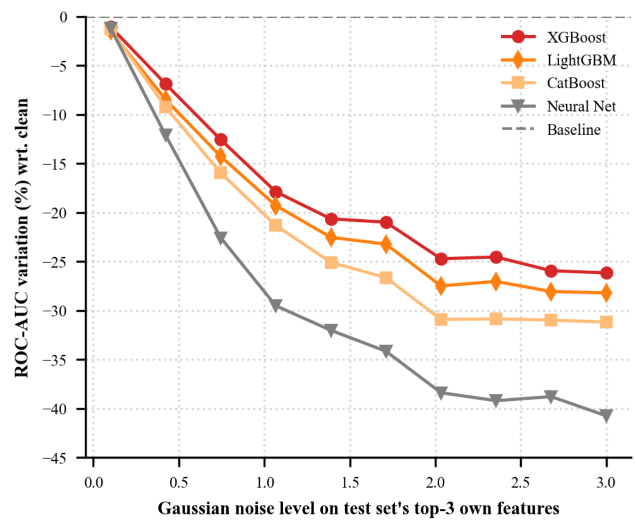
The results indicate a relatively good response from the tree based architectures, with the boosted models performing consistently better (as far as a $\sim 12\%$ at 1σ noise level) than the NN. While highly adaptable, neural architectures [13] are prone to overfitting; therefore, this accelerated degradation likely stems from the network relying on the specific, uncorrupted structure of the training data.

To further isolate feature dependency, we selectively target the test set's 3 most important features (according to each model's SHAP score) with analogous Gaussian noise (Figure 3b) in order to determine whether the models rely exclusively on primary features, or if they can pivot to secondary information when primary signals are corrupted. The results show how the latter is the case for boosted models, whose performance stops deteriorating at $\sim 2\sigma$, and the former for the NN, leading to an even bigger performance gap between the two types of models.

To further enhance generalization and preemptively harden these models against noisy test data, we also explore the effects of adding analogous noise in the training set (a.k.a. *data augmentation*); we do not observe, however, a statistically significant improvement in performance (but rather a generally worse one), neither by



(a) All features

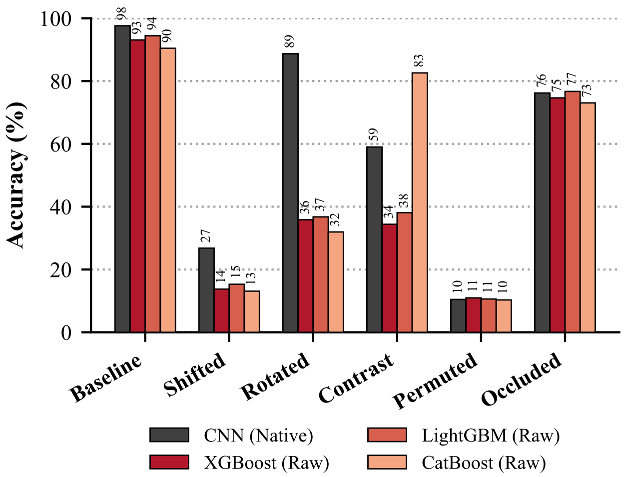


(b) 3 most important SHAP features

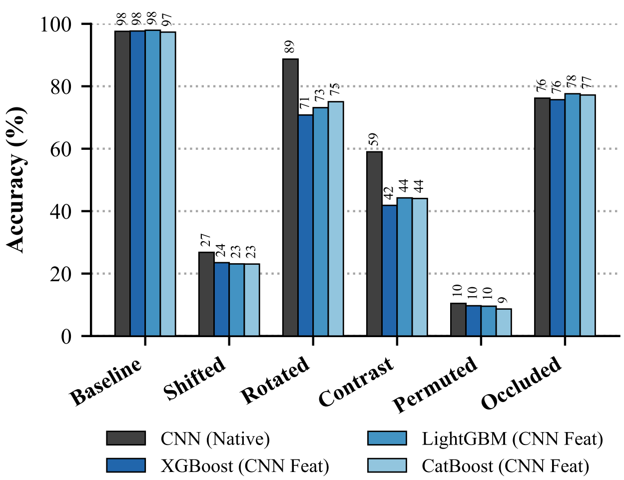
FIG. 3: Performance degradation under Gaussian noise injected into (a) all features and (b) only the 3 most important SHAP features. Values represent the shift in ROC-AUC relative to the baseline performance on clean data.

adding Gaussian noise to all features, nor by randomly flipping a portion of the labels, nor by simply reducing the dataset's size. Therefore, we omit those results from this discussion.

3 - Spatial Locality, Image Classification and Scalability in High Dimensional Spaces We evaluate the performance of the boosted models against a CNN baseline on the MNIST dataset [3] under various transformations, including rotation, translation, regional occlusion, complete loss of spatial coherence (pixel permutation), and contrast shifts. The results of this comparative analysis are presented in Figure 4a.



(a) Raw features



(b) Hybrid CNN-GBDT

FIG. 4: Comparison of predictive accuracy between boosted models (shades of red/blue) and a native CNN (gray) on the MNIST dataset under various spatial and morphological operations, for (a) raw features and (b) the hybrid CNN-GBDT pipeline.

While the boosted models achieve baseline accuracies comparable to the CNN demonstrating their strong classification capacity even in relatively high dimensional spaces their fundamental lack of spatial inductive biases becomes evident under geometric stress. The tree-based models suffer severe performance degradation when subjected to rototranslation operations, as they possess no inherent mechanism to account for spatial locality.

However, a notable exception occurs under contrast shifting, where CatBoost exhibits remarkable performance. This robustness is likely attributable to its symmetric oblivious tree structure and internal feature discretization algorithms, which inherently regularize the model against global scale transformations in the data.

To address this inherent lack of spatial inductive bias, we implemented a hybrid architecture. In this pipeline, a CNN operates as a spatial feature extractor, compressing the dataset into lower dimensional, dense vector representations that are subsequently used to train the boosted models.

The results of this hybrid approach are detailed in Figure 4b.

While the hybrid GBDT models do not uniformly surpass the native CNN across all perturbed domains, they surprisingly achieve superior accuracy on the unperturbed baseline images. Furthermore, this pipeline enables the tree based models to recover substantial predictive power on previously catastrophic tasks, most notably the classification of rotated images.

These findings demonstrate that while standalone decision trees struggle natively with geometrically complex data, integrating them into hybrid representation-learning pipelines can effectively bridge their structural deficits while leveraging their highly optimized classification boundaries.

Beyond geometric resilience, a critical constraint of gradient boosted architectures in high dimensional or large scale regimes is their computational efficiency. To evaluate this performance cost trade off, we analyzed the operational profiles of the models by contrasting their computational latency directly against their predictive performance, measured via the ROC-AUC score.

The efficiency dynamics of the three frameworks are illustrated in figure 5, where the NN is shown to achieve similar accuracy but with much higher latency than boosted models.

This trade off visualization reveals distinct algorithmic signatures for each framework, defining their effective Pareto frontiers. While all models naturally strive to maximize the ROC-AUC metric, the computational cost required to converge upon optimal performance varies significantly, as illustrated in the previous section.

4 - Temporal Dependencies, Sequential Forecasting and Multi-Classification

This analysis evaluates the *Human Activity Recognition Using Smartphones* (HAR) dataset, derived from experiments where volunteers wearing a smartphone on their wrist performed six distinct activities: walking, walking upstairs, walking downstairs, sitting, standing, and laying. The evaluation is structured into three experimental phases to assess how different data representations influence model performance:

- **Phase 1 (Raw Data)**: The originally 3D raw time series sensor signals are flattened into 2D vectors, and the models are trained directly on these representations. To determine the optimal combination of hyperparameters, the pipeline utilizes

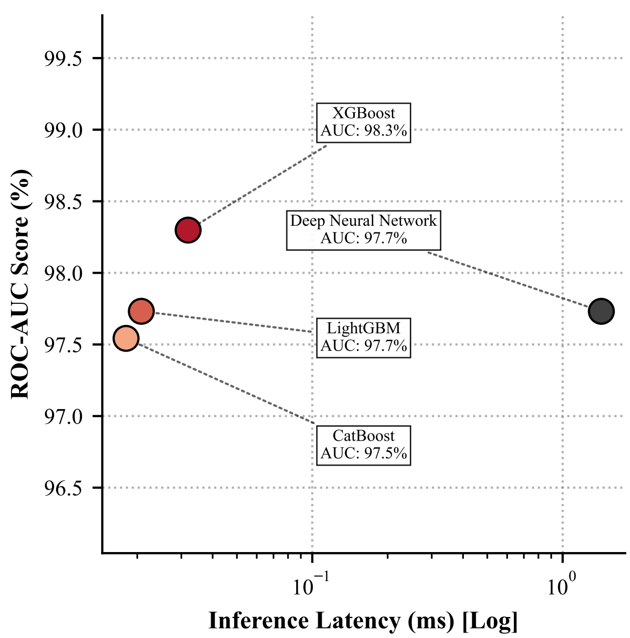


FIG. 5: Performance-efficiency trade-off on the MNIST set across the evaluated GBDT frameworks. The plot maps computational latency in classifying a given sample against the achieved ROC-AUC [14] score (%), highlighting the models' capacity to maximize predictive accuracy while minimizing execution cost.

`HalvingRandomSearchCV` [12].

- **Phase 2 (Enhanced Features)**: An `extract_enhanced_features` routine is applied to derive critical domain specific insights, transforming the continuous numerical streams into structured tabular columns optimized for decision trees. This engineering includes:
  - **Isolating Gravity**: The total acceleration recorded by a smartphone is a combination of the user's body movement and Earth's gravity. By subtracting the estimated body movement, the gravity vector is isolated, allowing the model to distinguish between vertical and horizontal postures.
  - **L2 Norm**: The magnitude of the 3D movement is collapsed into a singular scalar value.
  - **Time Domain Statistics**: Summary metrics (such as standard deviation and peak values) are computed over 128 frame sliding windows for all activities.
  - **Fast Fourier Transform (FFT, [21])**: Applied to extract frequency domain features, which capture the highly distinct, rhythmic patterns characteristic of specific activities.
- **Phase 3 (CNN Feature Extraction)**: A 1D Convolutional Neural Network (CNN) [13] is trained for 15 epochs on the raw temporal data to automatically recognize and extract abstract structural patterns. The resulting dense, refined representations are subsequently passed to XGBoost, LightGBM , and CatBoost for final classification.

In addition to the tree based architectures, the analysis includes a Recurrent Neural Network (RNN) baseline, comprising an LSTM [9] layer followed by a 128 neuron hidden layer. The training protocol involves a rigorous Grid Search combined with Early Stopping to determine the optimal hyperparameter configuration prior to final training.

The predictive performance of these diverse representations and architectures is detailed in Figure 6.

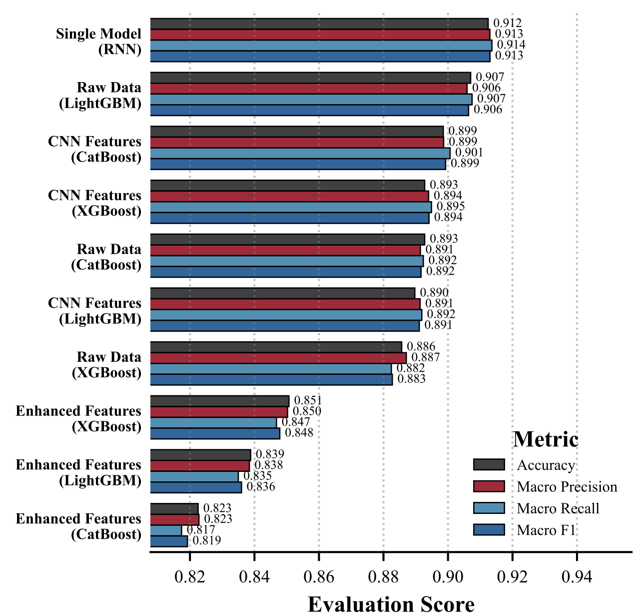


FIG. 6: Comparison of predictive performance metrics across various modeling pipelines and data representations on the HAR dataset [4]. Models are sorted in descending order of overall performance.

As expected, the LSTM [9] based RNN achieves the highest overall performance, yielding a metric score exceeding 0.91.

However, an unexpected outcome is that tree based models trained on the raw flattened data outperformed those trained on the manually engineered enhanced features. Notably, LightGBM trained on raw data secured the second-highest overall placement with an impressive score of ~ 0.907, whereas the models relying on domain specific feature engineering clustered at the very bottom of the rankings.

This might suggest that flattening the raw signals allowed the decision trees to identify latent sequential patterns that were inadvertently smoothed or discarded by the explicit mathematical aggregations applied in Phase 2.

Furthermore, the hybrid CNN GBDT pipelines exhibited highly stable and consistent performance, achieving strong scores between 0.890 and 0.901. This confirms that utilizing a NN for automated abstract feature extraction (Phase 3) yields a significantly superior data representation for tree-based classifiers compared to manual mathematical calculations.

Finally, for nearly every evaluated pipeline, all metrics used report virtually identical values. This indicates that the dataset is balanced and that the models perform consistently across all six activity classes.

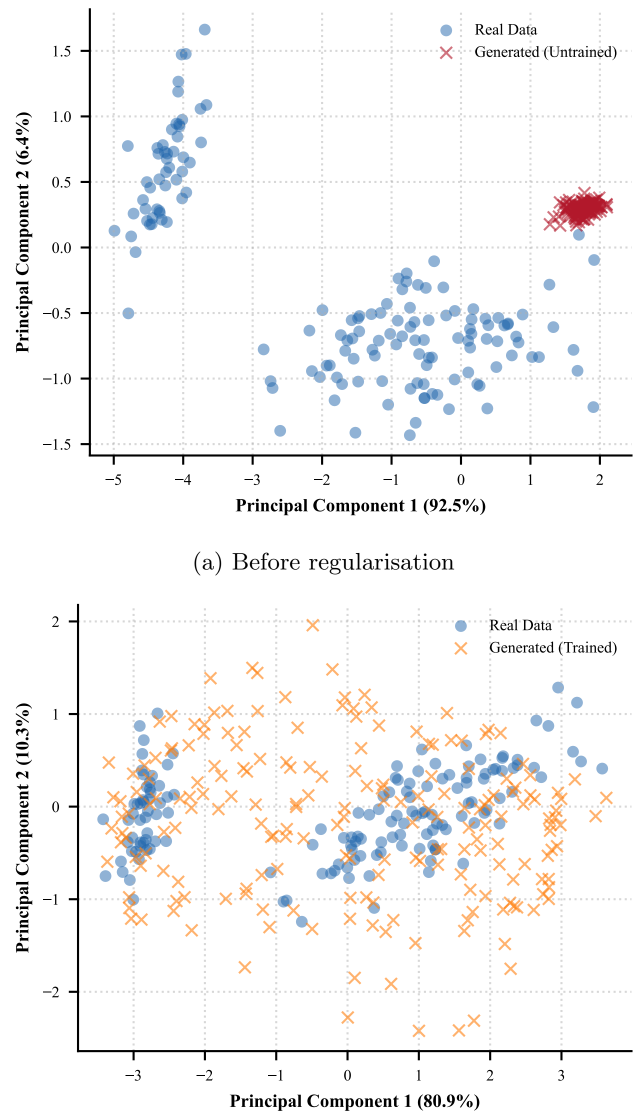
5 - Generative models based on boosting

Gradient boosted trees are highly optimized for rapid computation and robust classification, particularly on tabular datasets like the IRIS dataset. In this section, we extend their application from discriminative tasks to a generative environment by integrating them into a Generative Adversarial Network (GAN) architecture [22]. A standard GAN comprises two components: a generator that synthesizes data and a discriminator that distinguishes real from generated samples. We explore whether a GBDT discriminator can be integrated into a GAN framework to drive a neural generator, despite the non-differentiability of tree-based losses.

However, because the loss functions of tree-based models are non differentiable and cannot be backpropagated, we formulated the training process within a Reinforcement Learning (RL) framework [23]. In this paradigm, the standard discriminator loss is replaced by a policy gradient reward system that guides the neural generator based on the GBDT's evaluations.

The initial results of this baseline RL-GAN formulation are illustrated in Figure 7a.

As depicted, the base model exhibits severe structural pathologies, heavily overfitting to specific samples and experiencing complete mode collapse. To overcome these failure modes and stabilize the adversarial training process, several explicit regularization techniques were implemented to target the underlying sources of instability. To stabilize and optimize the training process, **L2 weight decay** [13] was applied to prevent overfitting by penalizing overly complex representations. An **entropy bonus** was utilized to discourage the generated data from concentrating in a narrow region, thereby actively favoring variance. Additionally, **noise injection** was introduced to prevent the generator from fixating on individual samples; this stochastic variability ensures the generator does not repeatedly produce identical outputs. Through **Z-score rescaling**, the received rewards are dynamically normalized so the generator does not prematurely converge on a "good enough" local optimum, but instead continuously optimizes its policy throughout training.



(a) Before regularisation

(b) After regularisation

FIG. 7: PCA projection of generated data from the RL-GAN framework (a) before and (b) after explicit regularisation (MMD, entropy bonus, weak critic). The baseline model suffers from severe mode collapse, while the regularised model blankets the target manifold without collapse.

Furthermore, because an overconfident discriminator often stifles generator learning in GAN models, we employed a **weak critic** strategy by deliberately restricting the structural capacity of the GBDT discriminator. Finally, we introduced a combination of **Maximum Mean Discrepancy (MMD) and repelling loss** [24] [25] to enforce spatial diversity. This combined objective forces the generated distribution to align with the empirical density of the original dataset while ensuring the synthetic examples remain distinct from one another. The

MMD, a robust kernel based method, was specifically chosen over standard moment matching, as the latter frequently resulted in bimodal clusters that matched base statistical moments but failed to capture the true continuous distribution.

The outcome of applying these combined regularization strategies is presented in Figure 7b, where a simple (non-quantitative) PCA analysis shows acceptable results.

CONCLUSIONS

In this work, we observed how on structured low-dimensional tabular data, both NN and boosted models can rapidly converge to a ceiling (Figure 1), though GBDTs demonstrated superior computational efficiency on moderately sized datasets. Furthermore, tree-based models exhibited an architectural resilience to data corruption and targeted feature noise (Figure 3), successfully pivoting to secondary signals where NNs succumbed to overfitting.

In unstructured domains (spatial and sequential data), GBDTs suffered severe degradation under geometric perturbations (Figure 4). However we demonstrated that hybrid pipelines utilizing CNNs for automated feature extraction paired with GBDT classifiers effectively bridge this structural deficit, frequently surpassing native deep learning baselines - though results may not generalize beyond this relatively simple dataset. Crucially, on time-series tasks, raw signal flattening and CNN-based extraction significantly outperformed manual domain specific feature engineering, which inadvertently destroyed latent sequential patterns (Figure 6).

Finally, we established that GBDTs can successfully drive continuous generative models (RL-GANs). While their sharp decision boundaries natively induce mode collapse, applying strict regularizations can stabilize the adversarial dynamic (Figure 7).

Declaration of AI: Gemini is used in order to refine the language and correct grammatical errors. Authors reviewed and edited the content as needed.

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